

M00 - Computer, Setup & Learning Foundations

8-lesson learner book | Beginner-first professional institute standard | B01-style teaching contract

Purpose

Become safe and independent with files, folders, software, VS Code, terminal use and study evidence.

Course boundary

This is the independent Standalone Data Engineer course, not Four-Course C3. Complete the course in sequence; each new idea is still taught from first principles.

Option B institute identity | Charcoal #1F2937 | Teal #14B8A6 | Soft Blue #3B82F6

This book is your teacher

Use one lesson at a time. Do not read protected review before saving a genuine attempt. Every lesson follows the same beginner-first sequence even when the engineering idea becomes more technical.

Daily lesson loop

1) why it matters -> 2) what you learn -> 3) prerequisite -> 4) picture it -> 5) new words -> 6) useful visual/source -> 7) follow once -> 8) expected result -> 9) why it worked -> 10) mistakes -> 11) recovery -> 12) independent task -> 13) validate -> 14) teach it back -> 15) protected review + changed retry -> 16) evidence/mastery + exact next route.

Prerequisite check

No prior technical background is required. You need a Windows PC, permission to create a working folder, and willingness to save evidence rather than guess.

Attempt protection

A protected review is repair material, not a shortcut. Save the first attempt, open only the matching review if needed, record the repair, close review, then solve a changed retry from a clean state.

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Every formal lesson and its actual page number in this exact rebuilt book.

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Prerequisite, tool and outcome map

ASSUMED

No prior technical background is required. You need a Windows PC, permission to create a working folder, and willingness to save evidence rather than guess.

MODULE PURPOSE

Become safe and independent with files, folders, software, VS Code, terminal use and study evidence.

EXIT EVIDENCE

Module practical + fresh retest + saved evidence; no critical unresolved failure.

Protected study route

Teaching -> assigned visual/source when useful -> follow once -> close worked material -> independent attempt -> validate -> save evidence -> teach it back. If weak: matching protected review -> targeted repair -> close review -> changed fresh retry -> revalidate.

DE00-01 - L01 - Files, Folders, Drives and Paths

1. Why it matters

Later tools fail when the learner cannot distinguish file identity from file location.

2. What you are learning

Build and explain the competency: L01 - Files, Folders, Drives and Paths.

3. Prerequisite / what you already know

No prior technical background is required. You need a Windows PC, permission to create a working folder, and willingness to save evidence rather than guess.

4. Picture it in your head

Think of a file as one document, a folder as a drawer, a drive as the cabinet, and the path as the full street address that tells Windows exactly where the document lives.

5. New words before we use them

A file name is like a person name; a path is the full address that tells the computer where the item lives.

6. Useful video / source

Source: Microsoft Support — File Explorer in Windows

Assigned scope: Files/folders, navigation, copy/move and organization

How to use it: Use this current Windows guide as a visual companion to the Teaching Book. Do the course folder/path task yourself; no handwritten notes.

OPEN VIDEO / SOURCE - CLICK HERE (support.microsoft.com)

URL: <https://support.microsoft.com/en-US/Windows/Experience/FileExplorer/file-explorer-in-windows>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

Create Documents\DE_COURSE_PRACTICE\files_lab, add location_test.txt, reopen File Explorer and locate it without Recent.

8. Expected result / what you should see

One stable course folder, one files_lab folder and one location_test.txt that remains after File Explorer is closed.

9. Why it worked

The setup action is trustworthy because the actual file/system state is checked after the operation; successful clicking or command execution alone is not treated as proof. Your independent check is: For L01 - Files, Folders, Drives and Paths: Verify the exact path, filename/extension, byte size or hash where appropriate, and confirm the original/working-copy rule was followed. The proof is the filesystem/system state, not the fact that a click or command completed.

10. Common beginner mistakes

- Looking in Downloads when the file is in Documents.
- Creating project_final_final folders instead of one stable workspace.
- Using Recent instead of knowing the actual location.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Think of a file as one document, a folder as a drawer, a drive as the cabinet, and the path as the full street address that tells Windows exactly where the document lives. 3. Run the earliest independent control: For L01 - Files, Folders, Drives and Paths: Verify the exact path, filename/extension, byte size or hash where appropriate, and confirm the original/working-copy rule was followed. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

Inside files_lab create practice/my_address_test.txt. Close File Explorer, reopen from Documents, find the file without Recent and write its full path inside the file. Evidence: Screenshot or note containing the exact folder path and the full file path.

13. Validate your result

For L01 - Files, Folders, Drives and Paths: Verify the exact path, filename/extension, byte size or hash where appropriate, and confirm the original/working-copy rule was followed.

14. Teach it back in your own words

Without notes, explain L01 - Files, Folders, Drives and Paths in your own words. Start with the purpose: Build and explain the competency: L01 - Files, Folders, Drives and Paths. Then name one realistic failure mode and explain how you would prove the result using this check: For L01 - Files, Folders, Drives and Paths: Verify the exact path, filename/extension, byte size or hash where appropriate, and confirm the original/working-copy rule was followed.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for L01 - Files, Folders, Drives and Paths with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-02.

DE00-02 - L02 - File Extensions and Common Data Files

1. Why it matters

Data engineering tools choose parsers from the real format, not from the icon shown in Windows.

2. What you are learning

Build and explain the competency: L02 - File Extensions and Common Data Files.

3. Prerequisite / what you already know

Prerequisite: DE00-01 mastered with saved evidence.

4. Picture it in your head

Picture two files with the same visible name but different labels on the box: .csv, .json, .py and .xlsx tell the computer what format is inside; changing the label does not convert the contents.

5. New words before we use them

The extension describes the file format. The app icon only shows a default way Windows may open it.

6. Useful video / source

Source: Microsoft Support — Common file name extensions in Windows

Assigned scope: Windows 10/11: what extensions mean, examples, showing extensions, and why renaming does not convert a format

How to use it: Use the current Microsoft guide beside the Teaching Book, then identify .csv/.json/.xlsx/.pdf/.py/.sql/.zip files on your own PC. No handwritten notes.

[OPEN VIDEO / SOURCE - CLICK HERE \(support.microsoft.com\)](https://support.microsoft.com/en-gb/windows/common-file-name-extensions-in-windows-da4a4430-8e76-89c5-59f7-1cdbbc75cb01)

URL:

<https://support.microsoft.com/en-gb/windows/common-file-name-extensions-in-windows-da4a4430-8e76-89c5-59f7-1cdbbc75cb01>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

Turn on File name extensions. Open sample.csv first in a text editor and then in Excel if available; observe that it remains CSV.

8. Expected result / what you should see

Extensions are visible and the same CSV can be inspected as comma-separated text and as rows/columns.

9. Why it worked

The setup action is trustworthy because the actual file/system state is checked after the operation; successful clicking or command execution alone is not treated as proof. Your independent check is: For L02 - File Extensions and Common Data Files: Verify the exact path, filename/extension, byte size or hash where appropriate, and confirm the original/working-copy rule was followed. The proof is the filesystem/system state, not the fact that a click or command completed.

10. Common beginner mistakes

- Renaming .csv to .xlsx and assuming conversion happened.
- Hiding extensions and creating report.csv.csv.
- Opening code files in rich-text software that can damage plain text.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Picture two files with the same visible name but different labels on the box: .csv, .json, .py and .xlsx tell the computer what format is inside; changing the label does not convert the contents. 3. Run the earliest independent control: For L02 - File Extensions and Common Data Files: Verify the exact path, filename/extension, byte size or hash where appropriate, and confirm the original/working-copy rule was followed. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

Choose four bundled files, predict each format from its extension before opening, then verify with a suitable app. Evidence: A four-row note: filename, predicted format, app used, what you observed.

13. Validate your result

For L02 - File Extensions and Common Data Files: Verify the exact path, filename/extension, byte size or hash where appropriate, and confirm the original/working-copy rule was followed.

14. Teach it back in your own words

Without notes, explain L02 - File Extensions and Common Data Files in your own words. Start with the purpose: Build and explain the competency: L02 - File Extensions and Common Data Files. Then name one realistic failure mode and explain how you would prove the result using this check: For L02 - File Extensions and Common Data Files: Verify the exact path, filename/extension, byte size or hash where appropriate, and confirm the original/working-copy rule was followed.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for L02 - File Extensions and Common Data Files with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-03.

DE00-03 - L03 - Download, ZIP, Extract, Copy, Move and Rename

1. Why it matters

Projects often arrive as archives. Keeping source and working copies separate prevents accidental evidence loss.

2. What you are learning

Build and explain the competency: L03 - Download, ZIP, Extract, Copy, Move and Rename.

3. Prerequisite / what you already know

Prerequisite: DE00-02 mastered with saved evidence.

4. Picture it in your head

Imagine a ZIP as a sealed travel bag. Extracting opens a working copy; copying keeps the original in place; moving changes location; renaming changes the label, not the contents.

5. New words before we use them

A ZIP is a sealed travel bag. Extraction unpacks normal working copies; the ZIP can remain as the clean source.

6. Useful video / source

Source: Microsoft Support — Zip and unzip files

Assigned scope: Create ZIP, Extract All, move/rename extracted files

How to use it: Follow the official Windows steps once, then perform the course ZIP/extract/copy/move/rename task without the guide.

OPEN VIDEO / SOURCE - CLICK HERE (support.microsoft.com)

URL: <https://support.microsoft.com/en-US/Windows/Experience/Storage-FileManagement/zip-and-unzip-files>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

Copy DOWNLOAD_ME_FIRST.zip to Downloads, Extract All, rename the extracted folder m00_download_lab and move that working folder into Documents\DE_COURSE_PRACTICE\01_MODULE_WORK\M00. Keep the original ZIP.

8. Expected result / what you should see

Both the source ZIP and a separate editable extracted working folder exist.

9. Why it worked

The setup action is trustworthy because the actual file/system state is checked after the operation; successful clicking or command execution alone is not treated as proof. Your independent check is: For L03 - Download, ZIP, Extract, Copy, Move and Rename: Verify the exact path, filename/extension, byte size or hash where appropriate, and confirm the original/working-copy rule was followed. The proof is the filesystem/system state, not the fact that a click or command completed.

10. Common beginner mistakes

- Editing inside the ZIP preview.
- Moving the only clean source when a copy was intended.
- Downloading duplicate copies instead of locating the first one.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Imagine a ZIP as a sealed travel bag. 3. Run the earliest independent control: For L03 - Download, ZIP, Extract, Copy, Move and Rename: Verify the exact path, filename/extension, byte size or hash where appropriate, and confirm the original/working-copy rule was followed. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

Copy messy_downloads to messy_downloads_working and organize only the working copy. Keep source untouched. Evidence: Before/after folder listing proving the source is unchanged and the working copy was reorganized.

13. Validate your result

For L03 - Download, ZIP, Extract, Copy, Move and Rename: Verify the exact path, filename/extension, byte size or hash where appropriate, and confirm the original/working-copy rule was followed.

14. Teach it back in your own words

Without notes, explain L03 - Download, ZIP, Extract, Copy, Move and Rename in your own words. Start with the purpose: Build and explain the competency: L03 - Download, ZIP, Extract, Copy, Move and Rename. Then name one realistic failure mode and explain how you would prove the result using this check: For L03 - Download, ZIP, Extract, Copy, Move and Rename: Verify the exact path, filename/extension, byte size or hash where appropriate, and confirm the original/working-copy rule was followed.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for L03 - Download, ZIP, Extract, Copy, Move and Rename with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-04.

DE00-04 - L04 - Installing Software Safely

1. Why it matters

Blind early installation creates unnecessary failures and version conflicts.

2. What you are learning

Build and explain the competency: L04 - Installing Software Safely.

3. Prerequisite / what you already know

Prerequisite: DE00-03 mastered with saved evidence.

4. Picture it in your head

Treat installation like accepting a tool from a supplier: verify the source, know what executable you launched, then confirm the tool works before adding more software.

5. New words before we use them

An installer delivers/configures an application. PATH is a search list a shell uses to find command programs.

6. Useful video / source

Source: Microsoft Support — Install programs from online sources

Assigned scope: Trusted publisher/download/install workflow

How to use it: Use this as the safety overlay. Install only the tool requested by the current module and verify the publisher/source first.

[OPEN VIDEO / SOURCE - CLICK HERE \(support.microsoft.com\)](https://support.microsoft.com/en-us/windows/apps/how-to-install-programs-from-online-sources-on-windows)

URL: <https://support.microsoft.com/en-us/windows/apps/how-to-install-programs-from-online-sources-on-windows>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

Using VS Code as the example, identify the official download page, the installed Start-menu application and (if still present) the installer file. Do not install unrelated tools.

8. Expected result / what you should see

Learner can explain installer vs application, source trust, version verification and PATH without installing the whole DE stack.

9. Why it worked

The setup action is trustworthy because the actual file/system state is checked after the operation; successful clicking or command execution alone is not treated as proof. Your independent check is: For L04 - Installing Software Safely: record the actual before/after state and verify the lesson requirement with an independent control. The proof is the filesystem/system state, not the fact that a click or command completed.

10. Common beginner mistakes

- Clicking sponsored download buttons.
- Approving UAC prompts without checking what launched them.
- Installing Python/PostgreSQL/WSL/Docker early "just in case".

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Treat installation like accepting a tool from a supplier: verify the source, know what executable you launched, then confirm the tool works before adding more software. 3. Run the earliest independent control: For L04 - Installing Software Safely: record the actual before/after state and verify the lesson requirement with an independent control. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

Pick one application already on the PC. Identify how you launch it and, if available, the old installer. Explain why they are different. Evidence: Short note with app name, launch location, installer location/status and verification method.

13. Validate your result

For L04 - Installing Software Safely: record the actual before/after state and verify the lesson requirement with an independent control. Challenge this expected outcome directly: Learner can explain installer vs application, source trust, version verification and PATH without installing the whole DE stack.

14. Teach it back in your own words

Without notes, explain L04 - Installing Software Safely in your own words. Start with the purpose: Build and explain the competency: L04 - Installing Software Safely. Then name one realistic failure mode and explain how you would prove the result using this check: For L04 - Installing Software Safely: record the actual before/after state and verify the lesson requirement with an independent control.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for L04 - Installing Software Safely with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-05.

DE00-05 - L05 - VS Code From Zero

1. Why it matters

Opening the correct root makes paths, terminals and later project structure understandable.

2. What you are learning

Build and explain the competency: L05 - VS Code From Zero.

3. Prerequisite / what you already know

Prerequisite: DE00-04 mastered with saved evidence.

4. Picture it in your head

VS Code is your workbench: the project folder is the bench, files are parts laid out on it, and the terminal is a tool drawer attached to the same workspace.

5. New words before we use them

VS Code is a technical workspace/desk; it is not Python, SQL, Git or a database.

6. Useful video / source

Source: Visual Studio Code — Official beginner tutorial

Assigned scope: 0:00–7:00

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[OPEN VIDEO / SOURCE - CLICK HERE \(youtube.com\)](https://www.youtube.com/watch?v=B-s71n0dHUK)

URL: <https://www.youtube.com/watch?v=B-s71n0dHUK>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

Open m00_download_lab as a folder, inspect data and notes, create evidence.txt at project root, save, close and reopen.

8. Expected result / what you should see

Explorer shows the whole project tree and evidence.txt survives in the intended location.

9. Why it worked

The setup action is trustworthy because the actual file/system state is checked after the operation; successful clicking or command execution alone is not treated as proof. Your independent check is: For L05 - VS Code From Zero: record the actual before/after state and verify the lesson requirement with an independent control. The proof is the filesystem/system state, not the fact that a click or command completed.

10. Common beginner mistakes

- Opening one random file instead of the project root.
- Creating a file in the wrong selected subfolder.
- Closing an Untitled/unsaved tab.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: VS Code is your workbench: the project folder is the bench, files are parts laid out on it, and the terminal is a tool drawer attached to the same workspace. 3. Run the earliest independent control: For L05 - VS Code From Zero: record the actual before/after state and verify the lesson requirement with an independent control. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

Open supplier_import_assets as project root. Create evidence/hello.txt at the correct level, close and reopen, then verify location. Evidence: Project-tree screenshot or written path to evidence/hello.txt plus one sentence explaining project root.

13. Validate your result

For L05 - VS Code From Zero: record the actual before/after state and verify the lesson requirement with an independent control. Challenge this expected outcome directly: Explorer shows the whole project tree and evidence.txt survives in the intended location.

14. Teach it back in your own words

Without notes, explain L05 - VS Code From Zero in your own words. Start with the purpose: Build and explain the competency: L05 - VS Code From Zero. Then name one realistic failure mode and explain how you would prove the result using this check: For L05 - VS Code From Zero: record the actual before/after state and verify the lesson requirement with an independent control.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for L05 - VS Code From Zero with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-06.

DE00-06 - L06 - Terminal Fundamentals

1. Why it matters

Many later DE tools are command-driven; location awareness prevents most beginner terminal mistakes.

2. What you are learning

Build and explain the competency: L06 - Terminal Fundamentals.

3. Prerequisite / what you already know

Prerequisite: DE00-05 mastered with saved evidence.

4. Picture it in your head

Picture the terminal as a robot that follows exactly one typed instruction in the current folder. The same command can affect different files if you stand in a different directory.

5. New words before we use them

File Explorer uses clicks; a terminal uses typed instructions. Every command runs from a current directory.

6. Useful video / source

Source: VS Code terminal orientation — Tech With Tim

Assigned scope: 4:40–6:40

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[OPEN VIDEO / SOURCE - CLICK HERE \(youtube.com\)](https://www.youtube.com/watch?v=ORrELERGIHs)

URL: <https://www.youtube.com/watch?v=ORrELERGIHs>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

From the VS Code terminal at m00_download_lab: pwd, ls, cd data, pwd, ls, cd ., mkdir terminal_created.

8. Expected result / what you should see

Learner can predict where the new folder appears and can explain each path change.

9. Why it worked

The setup action is trustworthy because the actual file/system state is checked after the operation; successful clicking or command execution alone is not treated as proof. Your independent check is: For L06 - Terminal Fundamentals: record the actual before/after state and verify the lesson requirement with an independent control. The proof is the filesystem/system state, not the fact that a click or command completed.

10. Common beginner mistakes

- Running the right command from the wrong directory.
- Treating a path as if it were a command.
- Copying destructive commands without understanding them.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Picture the terminal as a robot that follows exactly one typed instruction in the current folder. 3. Run the earliest independent control: For L06 - Terminal Fundamentals: record the actual before/after state and verify the lesson requirement with an independent control. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

From supplier_import_assets root, enter data, list files, return to root, create archive, enter it, create 2026, and record pwd at each level. Evidence: Saved terminal transcript or screenshot sequence showing the path changes and created folders.

13. Validate your result

For L06 - Terminal Fundamentals: record the actual before/after state and verify the lesson requirement with an independent control. Challenge this expected outcome directly: Learner can predict where the new folder appears and can explain each path change.

14. Teach it back in your own words

Without notes, explain L06 - Terminal Fundamentals in your own words. Start with the purpose: Build and explain the competency: L06 - Terminal Fundamentals. Then name one realistic failure mode and explain how you would prove the result using this check: For L06 - Terminal Fundamentals: record the actual before/after state and verify the lesson requirement with an independent control.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for L06 - Terminal Fundamentals with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-07.

DE00-07 - L07 - Errors and Beginner Troubleshooting

1. Why it matters

Random multi-change troubleshooting hides the real cause and makes the next failure harder.

2. What you are learning

Build and explain the competency: L07 - Errors and Beginner Troubleshooting.

3. Prerequisite / what you already know

Prerequisite: DE00-06 mastered with saved evidence.

4. Picture it in your head

An error is a dashboard warning light, not a verdict. Read the first useful signal, isolate the smallest failing step, change one thing, then rerun the same check.

5. New words before we use them

An error is evidence. Symptom, cause and fix are different things.

6. Useful video / source

Internal teaching route

Internal teaching route. This lesson is fully taught in this book and its bundled practice. Use the mental model and worked demonstration here; no filler external video is required for mastery.

7. Follow with me once

Use exercises/broken_setup/TASK.txt. Read the symptom, find the misplaced file, repair only the location problem and write repair_note.txt while preserving broken_setup_source.

8. Expected result / what you should see

One controlled fix with a repair note that names symptom, cause, change and verification.

9. Why it worked

This works because build and explain the competency: L07 - Errors and Beginner Troubleshooting is made explicit before the result is accepted. Your independent check is: For L07 - Errors and Beginner Troubleshooting: record the actual before/after state and verify the lesson requirement with an independent control. The proof is the filesystem/system state, not the fact that a click or command completed.

10. Common beginner mistakes

- Reinstalling software to fix a path problem.
- Changing several settings at once.
- Deleting the broken original evidence.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: An error is a dashboard warning light, not a verdict. 3. Run the earliest independent control: For L07 - Errors and Beginner Troubleshooting: record the actual before/after state and verify the lesson requirement with an independent control. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

Use exercises/path_problem/TASK.txt. Save attempt note first, form one hypothesis at a time, fix the path issue and write a two-sentence cause/repair explanation. Evidence: attempt_note.txt plus repaired state plus explanation of cause and verification.

13. Validate your result

For L07 - Errors and Beginner Troubleshooting: record the actual before/after state and verify the lesson requirement with an independent control. Challenge this expected outcome directly: One controlled fix with a repair note that names symptom, cause, change and verification.

14. Teach it back in your own words

Without notes, explain L07 - Errors and Beginner Troubleshooting in your own words. Start with the purpose: Build and explain the competency: L07 - Errors and Beginner Troubleshooting. Then name one realistic failure mode and explain how you would prove the result using this check: For L07 - Errors and Beginner Troubleshooting: record the actual before/after state and verify the lesson requirement with an independent control.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for L07 - Errors and Beginner Troubleshooting with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-08.

DE00-08 - L08 - Study Evidence, Attempts and Review Discipline

1. Why it matters

Job work and interviews require recall and reasoning without an answer sheet.

2. What you are learning

Build and explain the competency: L08 - Study Evidence, Attempts and Review Discipline.

3. Prerequisite / what you already know

Prerequisite: DE00-07 mastered with saved evidence.

4. Picture it in your head

Use a lab notebook mental model: first attempt, observed result, repair, and changed retry are separate entries. If you erase the failed attempt, you lose the evidence of learning.

5. New words before we use them

Recognition is not recall. Preserve the first attempt because it shows what you actually knew before review.

6. Useful video / source

Internal teaching route

Internal teaching route. This lesson is fully taught in this book and its bundled practice. Use the mental model and worked demonstration here; no filler external video is required for mastery.

7. Follow with me once

Create 02_ATTEMPTS, 03_REVIEW, 04_RETESTS, 05_EVIDENCE. Save m00_test_attempt.txt, then a separate m00_test_retest.txt and evidence note.

8. Expected result / what you should see

Original attempt remains unchanged; retest is separate; evidence identifies both.

9. Why it worked

The query is trustworthy because the business question, row grain and transformation stages are explicit, then an independent row/key/total check challenges a plausible-but-wrong result. Your independent check is: For L08 - Study Evidence, Attempts and Review Discipline: Run an independent control query for row count, key coverage and one important total; inspect a deliberately changed filter/key case to prove the query logic rath. The proof is the filesystem/system state, not the fact that a click or command completed.

10. Common beginner mistakes

- Editing the first attempt after seeing review.
- Calling Reviewed the same as Mastered.
- Calling Skipped the same as Mastered.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Use a lab notebook mental model: first attempt, observed result, repair, and changed retry are separate entries. 3. Run the earliest independent control: For L08 - Study Evidence, Attempts and Review Discipline: Run an independent control query for row count, key coverage and one important total; inspect a deliberately changed filter/key case to prove the query logic rather than memorization 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

Create attempts/P05_attempt1.txt with an intentionally wrong sentence, then retests/P05_retest1.txt with correction; explain both in evidence/P05_evidence.txt. Evidence: Three separate files proving attempt, retest and explanation. M00 move-on standard Move to M01 only when you can find the course workspace, identify common file formats, extract and preserve source archives, explain safe installation, open a project root in VS Code, navigate a terminal safely, troubleshoot one path problem with one hypothesis at a time, and preserve attempts/retests/evidence separately.

13. Validate your result

For L08 - Study Evidence, Attempts and Review Discipline: Run an independent control query for row count, key coverage and one important total; inspect a deliberately changed filter/key case to prove the query logic rather than memorization.

14. Teach it back in your own words

Without notes, explain L08 - Study Evidence, Attempts and Review Discipline in your own words. Start with the purpose: Build and explain the competency: L08 - Study Evidence, Attempts and Review Discipline. Then name one realistic failure mode and explain how you would prove the result using this check: For L08 - Study Evidence, Attempts and Review Discipline: Run an independent control query for row count, key coverage and one important total; inspect a deliberately changed filter/key case to prove the query logic rath.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for L08 - Study Evidence, Attempts and Review Discipline with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: module practical / Gate route after the mastery checklist.